DAY 16 — ADVANCED PIVOTTABLES

Excel Data Analysis Training Notes

Day 16 Learning Objectives

By the end of Day 16, I can:

1. Group dates and numerical values in PivotTables.
2. Create Calculated Fields for business metrics.
3. Sort PivotTable results by a specific value field.
4. Apply Label Filters and Value Filters.
5. Perform Top/Bottom Analysis to identify business performance.

LESSON 1: GROUPING

1. What Is Grouping?

Grouping means combining individual values into meaningful categories to simplify analysis.

Instead of analyzing every transaction separately, I can summarize data by month, quarter, year, or numerical ranges.

2. Types of Grouping

A. Date Grouping

Date fields can be grouped by:

- Days
- Months
- Quarters
- Years

Example:

Month	Total Bookings
January	270
February	280
March	200
Grand Total	750

Business purpose: Identify monthly trends and compare performance across different periods.

B. Numeric Grouping

Numeric grouping combines numerical values into ranges.

Example:

Booking Value Range	Number of Bookings
₹0–₹1,999	2
₹2,000–₹3,999	2
₹4,000–₹5,999	1
₹6,000–₹7,999	2
₹8,000–₹9,999	1

Business purpose: Understand the distribution of transaction values, customer spending, or sales amounts.

3. How to Group Dates

1. Create a PivotTable.
2. Drag Date into Rows.
3. Drag Bookings into Values.
4. Right-click a date.
5. Select Group.
6. Choose Months and, where necessary, Years.
7. Click OK.

Important: When data covers multiple years, grouping by both Years and Months prevents January from different years from being combined unintentionally.

4. How to Group Numeric Values

1. Drag the numerical field into Rows.
2. Drag Transaction ID into Values and select Count.
3. Right-click a numerical row label.
4. Select Group.
5. Enter the starting value, ending value, and interval size.
6. Click OK.

Important: Ensure numerical fields contain valid numbers and date fields contain valid dates. Blank or invalid entries can prevent grouping.

LESSON 2: CALCULATED FIELDS

1. What Is a Calculated Field?

A Calculated Field creates a new calculation within a standard PivotTable using existing source fields.

It is useful when the source data contains Revenue and Cost but does not contain Profit.

2. Important Business Formulas

Profit

Profit = Revenue − Cost

Example:

Revenue = ₹80,000
Cost = ₹50,000

Profit = ₹80,000 − ₹50,000 = ₹30,000

Profit Margin

Profit Margin = (Profit ÷ Revenue) × 100

Example:

Profit = ₹30,000
Revenue = ₹80,000

Profit Margin = (₹30,000 ÷ ₹80,000) × 100 = 37.5%

3. Business Example

Country	Revenue	Cost	Profit	Profit Margin
India	₹80,000	₹50,000	₹30,000	37.5%
Kenya	₹60,000	₹35,000	₹25,000	41.7%
Nigeria	₹100,000	₹75,000	₹25,000	25.0%

Business interpretation:

- Highest revenue: Nigeria — ₹100,000.
- Highest profit amount: India — ₹30,000.
- Highest profit margin: Kenya — approximately 41.7%.

Important lesson: The country with the highest revenue does not necessarily have the highest profit or profit margin.

4. How to Create a Calculated Field

1. Create a PivotTable.
2. Add Country to Rows.
3. Add Revenue and Cost to Values.
4. Select a cell inside the PivotTable.
5. Go to PivotTable Analyze.
6. Select Fields, Items & Sets, then Calculated Field.
7. Enter the name Profit.
8. Enter the formula =Revenue-Cost using the available source field names.
9. Click Add or OK, depending on the Excel version.

5. Important Limitations

- A standard PivotTable Calculated Field calculates using source fields and then summarizes the results.
- Ratios may require special care. For an overall profit margin, divide total profit by total revenue.
- A simple average of country-level profit margins is not necessarily the overall profit margin.
- Calculated Fields in standard PivotTables differ from measures in the Excel Data Model.

LESSON 3: SORTING

1. What Is Sorting?

Sorting arranges data in a particular order to identify the highest or lowest values.

Descending Order

Largest to Smallest.

Example: ₹100,000 → ₹60,000 → ₹25,000

Ascending Order

Smallest to Largest.

Example: ₹25,000 → ₹60,000 → ₹100,000

2. Sorting by a Specific Value Field

Suppose a PivotTable contains Country, Bookings, Revenue, and Profit.

I must select the appropriate value field depending on the business question.

- Highest bookings: Sort by Bookings.
- Highest revenue: Sort by Revenue.
- Highest profit: Sort by Profit.
- Lowest profit: Sort by Profit from Smallest to Largest.

3. How to Sort a PivotTable

1. Click a numerical cell in the value field I want to analyze.
2. Right-click and select Sort.
3. Select Largest to Smallest or Smallest to Largest.
4. Check that the categories remain aligned with their values.

Business example:

Country	Profit
Tanzania	₹45,000
India	₹30,000
Kenya	₹25,000
Nigeria	₹25,000
Uganda	₹15,000

Sorting Profit from Largest to Smallest places Tanzania first.

Important: Sorting changes the display order, not the underlying values.

LESSON 4: FILTERING

1. What Is Filtering?

Filtering displays only the records or categories that meet selected conditions.

Filtering does not delete the underlying source data.

2. Label Filter

A Label Filter uses category names or text conditions.

Examples:

- Show India and Kenya only.
- Show countries beginning with N.
- Show product names containing Premium.

How to Apply a Label Filter

1. Open the dropdown beside the Row Labels field.
2. Select Label Filters.
3. Choose the required condition.
4. Enter the text or select the appropriate categories.
5. Click OK.

3. Value Filter

A Value Filter uses numerical values summarized in the PivotTable.

Examples:

- Profit greater than ₹25,000.
- Bookings greater than 400.
- Revenue below ₹50,000.

How to Apply a Value Filter

1. Open the dropdown beside the row labels.
2. Select Value Filters.
3. Choose the required condition.
4. Select the relevant value field, such as Sum of Profit.
5. Enter the condition.
6. Click OK.

4. Label Filter vs. Value Filter

Label Filter:

- Based on category names or text.
- Example: Country begins with N.
- Result: Nigeria.

Value Filter:

- Based on summarized numerical values.
- Example: Sum of Profit greater than ₹25,000.
- Result: India and Tanzania when their profits are ₹30,000 and ₹45,000, respectively.

Important: Greater Than ₹25,000 excludes values equal to ₹25,000. Use Greater Than or Equal To when equality should qualify.

5. Sorting vs. Filtering

Sorting changes the order of displayed results.

Filtering restricts the displayed results to those meeting a condition.

These operations solve different business questions.

LESSON 5: TOP/BOTTOM ANALYSIS

1. What Is Top/Bottom Analysis?

Top/Bottom Analysis identifies the highest- or lowest-performing categories using a selected numerical measure.

Examples:

- Top 3 countries by revenue.
- Bottom 2 branches by profit.
- Top 5 products by bookings.
- Bottom 10 salespeople by sales.

2. How to Apply a Top/Bottom Filter

1. Create a PivotTable with categories in Rows and the required metric in Values.
2. Open the dropdown beside the row labels.
3. Select Value Filters, then Top 10.
4. Choose Top or Bottom.
5. Enter the required number of items.
6. Select Items.
7. Choose the appropriate measure, such as Sum of Revenue or Sum of Profit.
8. Click OK.
9. Sort the results if a ranked display is required.

Note: Excel's Top 10 dialog allows a different number of items; it does not have to remain at 10.

3. Business Example: Top 3 Countries by Profit

Country	Profit
Tanzania	₹45,000
Ghana	₹35,000
India	₹30,000
Kenya	₹25,000
Nigeria	₹25,000
Uganda	₹15,000
Rwanda	₹10,000

Top 3 by Sum of Profit:

1. Tanzania — ₹45,000
2. Ghana — ₹35,000
3. India — ₹30,000

4. Business Example: Bottom 2 Countries by Revenue

Country	Revenue
Rwanda	₹30,000
Uganda	₹40,000

Bottom 2 by Sum of Revenue identifies Rwanda and Uganda.

5. Important Analyst Considerations

- Always select the correct metric before applying a Top/Bottom filter.
- Top 3 by Revenue can produce a different result from Top 3 by Profit.
- Ties at the cutoff can affect how the ranking is interpreted.
- A low-performing category requires investigation; it does not automatically mean poor management.
- Consider business targets, historical trends, transaction volume, costs, and operating conditions before making recommendations.

DAY 16 — FINAL REVISION CHEAT SHEET

Business Requirement	Excel Feature
Summarize bookings by month	Date Grouping
Combine booking values into ranges	Numeric Grouping
Calculate Profit within a PivotTable	Calculated Field
Find the highest revenue country	Sort Revenue Largest to Smallest
Find the lowest profit country	Sort Profit Smallest to Largest
Show only selected countries	Label Filter or item selection
Show countries with profit above ₹25,000	Value Filter
Find the Top 3 countries by profit	Top 3 by Sum of Profit
Find the Bottom 2 countries by revenue	Bottom 2 by Sum of Revenue

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KEY LESSONS FROM DAY 16

1. Grouping simplifies detailed data into meaningful categories.
2. Calculated Fields help create metrics such as Profit.

3. Profit and Profit Margin are different measures.
4. Sorting helps rank data according to a selected value field.
5. Label Filters work with category labels, while Value Filters work with numerical summaries.
6. Top/Bottom Analysis identifies high- and low-performing categories.
7. The chosen field or metric must match the manager's business question.
8. PivotTable results support investigation, but business conclusions require context.

Final Data Analyst Checklist

Before presenting a PivotTable analysis, I should ask:

- Have I grouped dates and values appropriately?
- Are the calculations correct?
- Am I sorting by the correct measure?
- Have I applied the correct filter?
- Have I selected the correct metric for Top/Bottom Analysis?
- Have I interpreted the results correctly?
- Do I need further drill-down into branches, products, salespeople, or time periods?

Day 16 Status: COMPLETED

Next learning objective: Continue building practical Excel analysis skills using business scenarios, PivotTable interpretation, and further drill-down analysis.